

VTPEH 6270 - Check Point 03

Basic R Skills

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1 Introduction

- Background

There is a growing proportion of parents of young children, who hesitate or refuse to vaccinate their children in the United States. Several immunizations are state-mandated for children in public schools. Among those are Tetanus, Diphtheria, Pertussis, Hepatitis B, Measles, Mumps, Rubella, Varicella. Although all states allow children to be exempted from immunization for medical reasons, some states are more stringent than

others in allowing exemptions to immunization for philosophical or religious reasons. Under the leadership of the new secretary of health and human services, R. F. Kennedy Jr., federal guidelines for recommended childhood immunizations are shifting and there are consequences in the stringency of immunization mandates for children attending public schools at the level of states. In a previous analysis, we found that the highest level of immunization resistance in 2024 is found in the state of Idaho reaching 15%.

2 Project Question

We will focus on the state of **Idaho** and try to find correlations between attitudes towards immunization in childhood and social determinants such as education levels, median income, poverty rates, population density, party affiliation, language isolation by comparing data at the level of counties, based on the hypothesis that immunization resistance is linked to party affiliation (Iasulaitis (2025)) and/or lack of education (Vardavas (2023)). I do not yet have access to immunization resistance data at the level of counties but have requested it from the Idaho Department of Health. In the meantime, I will look at the rates of voter registration in fringe political parties, such as the libertarian party and the Constitution party, and also look if correlations can be seen between education levels and median income (Edwards (2024)). As a control, to test the method, I will look if I can see an expected correlation between Median Income and Below Poverty Percentage at the scale of Idaho Counties.

3 Data Sets Overview

For now, we use the *Vaccination Coverage among Kindergartners* data set available at [https://data.cdc.gov/Vaccinations/Vaccination-Coverage-and-Exemptions-among-Kinderga/ijqb-a7ye/about_data]. The population studied is that of children enrolled in Kindergarten in public schools in the United states. The study reports on vaccination and exemption rates for the school years spanning 2009-2010 to 2024-2025. The data was collected mainly through the census report, which is a mandated collection of data reported by all public school administrators. We will also use voter registration data as of January 2025 from the website <https://voteidaho.gov/data-and-dashboards/voter-registration-totals/>. The data for Idaho poverty rates, median income, language isolation, education levels, population are collected for the year 2019-2023 and are available at <https://hdpulse.nimhd.nih.gov>.

```
# Set working directory
# C:\Users\alexm\OneDrive\Documents\CLASSES\VTPEH6270\Immunizations_K
setwd("C:/Users/alexm/REPOS/CRANE/VaccinationHesitancy/analysis")

# Load file "Immunization_K_Idaho" from excel file
# subset extracted from CDC "Vaccination_Coverage_and_Exemptions_among_Kindergartners_20260201"
data_K <-
  read.csv("../data/Immunizations_K_Idaho.csv", header=TRUE)
```

4 Data for Kindergarten Exemptions from Immunization in Idaho

The data set contains data that is either redundant or unnecessary for the analysis planned at this time. We create a new data frame after removing this data that we do not deem necessary at this time. We will go back to the original data set for following analysis.

```
# remove the columns geography_type, footnotes & survey_type

cleaned_data_K <-
```

```
data_K[, !names(data_K) %in%
        c("geography_type", "footnotes", "survey_type")]
```

```
library(ggplot2)
```

```
data_polio <- subset(cleaned_data_K, vaccine_exemption == "Polio")
```

```
ggplot(data_polio,
       aes( x = year, y = population_size )) +
  geom_col(fill = "lightblue") +
  labs( caption = "Figure 1. Population size surveyed for Polio immunization/exemption",
        x = "Year",
        y = "Population size") +
  scale_x_continuous(breaks = c(2009, 2014, 2019, 2024),
                    labels = c("2009", "2014", "2019", "2024"))
```

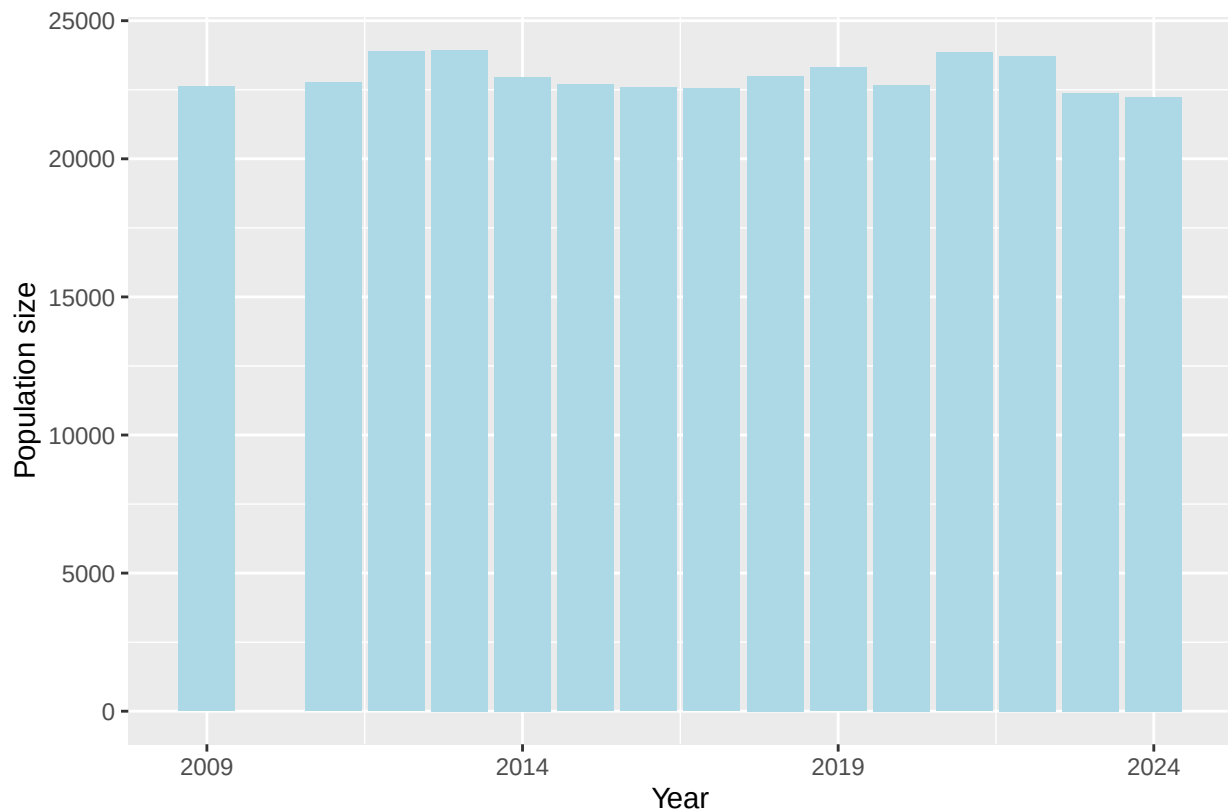


Figure 1. Population size surveyed for Polio immunization/exemption

```
# find years of study and mean population size over the years of study
```

```
min.year <- min(cleaned_data_K$school_year)
max.year <- max(cleaned_data_K$school_year)
mean.K.pop <- round(mean(data_polio$population_size),0)
print(mean.K.pop)
```

The time period covered by the study is 2009-10 to 2024-25 and the mean number of Kindergartners surveyed for Polio immunization status over the years is 2.3011×10^4 . We see that **data is missing for**

the school year 2010-2011.

To check for data completion, a Table for Vaccination Exemptions per school year is compiled.

```
# subset the data for exemption data only
data_exemptions <- subset(cleaned_data_K, vaccine_exemption == "Exemption")

library(dplyr)

table_exemptions <- data_exemptions %>%
  group_by(school_year) %>%
  summarise(exemption = sum(number_of_exemptions, na.rm = TRUE))

# make a nice table for the data
library(knitr)

kable(table_exemptions,
      col.names = c("School Year", "Total Exemptions"),
      caption = "Total Exemptions by School Year")
```

Table 1: Total Exemptions by School Year

School Year	Total Exemptions
2009-10	1720
2011-12	2452
2012-13	2798
2013-14	3080
2014-15	3000
2015-16	2778
2016-17	2956
2017-18	3194
2018-19	3556
2019-20	3528
2020-21	3722
2021-22	4669
2022-23	5718
2023-24	6384
2024-25	6854

The table shows that the data on number of exemptions granted in Idaho is available for all years of the study. Below is the plot showing the estimated vaccination exemption rate across school years for Idaho.

```
# plot the data after installing the ggplot2 package
library(ggplot2)

ggplot(data_exemptions,
      aes(year, estimate, color=dose)) +
  geom_point(size=3, alpha=0.5) +
  theme(plot.title = element_text(hjust=0.5, size=10)) +
  theme(plot.subtitle = element_text(hjust=0.5, size=10)) +
```

```
labs( caption = "Figure 2. Vaccination exemptions rates for kindergartners",
      subtitle = "(CDC data for Idaho State)",
      x = "Year",
      y = "Vaccination Exemption Rates (%) ",
      color = "Type of Exemption") +
scale_x_continuous(breaks = c(2009, 2014, 2019, 2024),
                  labels = c("2009", "2014", "2019", "2024")) +
geom_line()
```

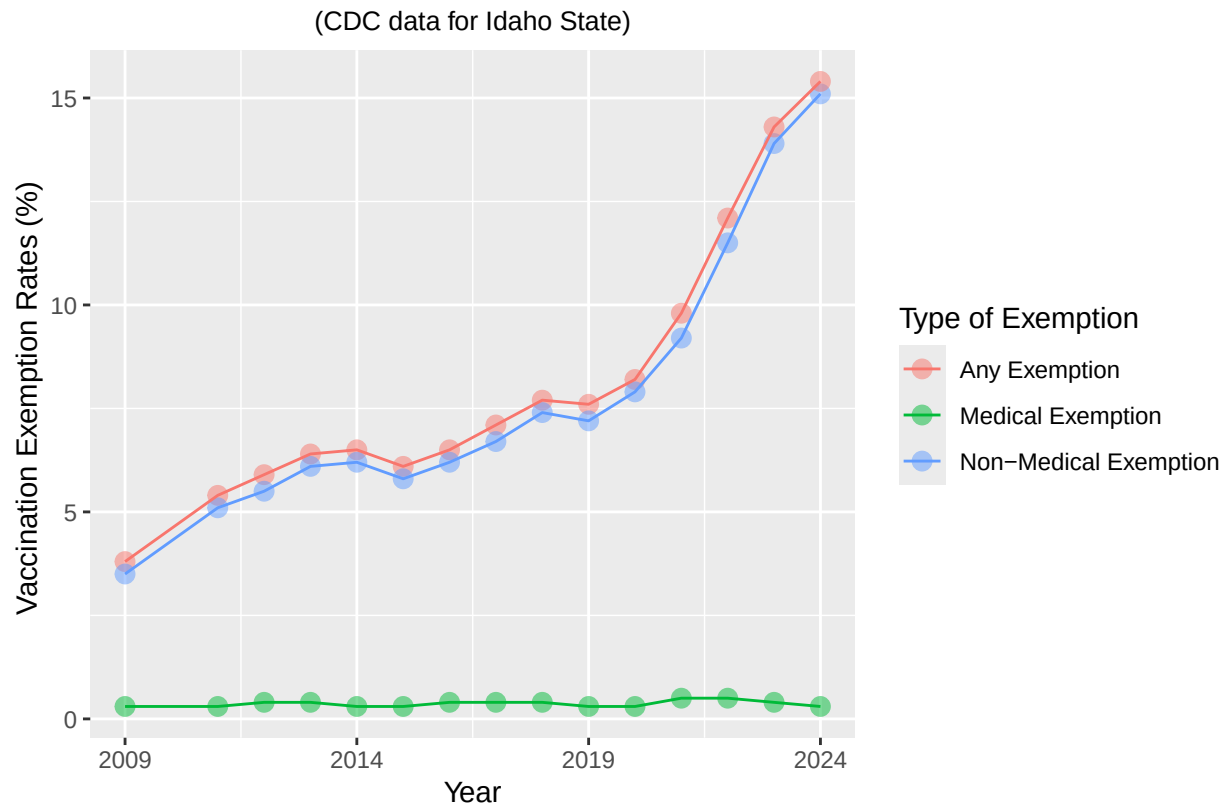


Figure 2. Vaccination exemptions rates for kindergartners

```
# find min and max of exemption rate
min_exemption <- min(subset(data_exemptions, dose == "Any Exemption")$estimate, na.rm = TRUE)
max_exemption <- max(subset(data_exemptions, dose == "Any Exemption")$estimate, na.rm = TRUE)

min_year <- (data_exemptions$year[which.min(subset(data_exemptions, dose == "Any Exemption")$estimate)])
max_year <- (data_exemptions$year[which.max(subset(data_exemptions, dose == "Any Exemption")$estimate)])
```

What do these plots show?

- This plot shows that the vaccination exemption rate in Idaho has steadily increased from a low of 3.8% since data was first collected in 2009, to reach its highest level 15.4% in 2024, which is the last year for which data is available.
- In contrast, the vaccination exemption rate for medical reasons has been relatively stable at rates under 0.5

Below is the plot showing the estimated vaccination rates per vaccine type across years for Idaho. We can see that all immunization types have decreased but DTP, Polio and Varicella more than MMR and Hepatitis. Decreased rates in Polio and Tetanus immunizations are particularly worrisome.

```
library(ggplot2)

data_vaccines <- subset(cleaned_data_K, vaccine_exemption != "Exemption")

ggplot(data_vaccines,
       aes(year, estimate, color=vaccine_exemption)) +
  geom_point(size=3, alpha=0.5) +
  theme(plot.title = element_text(hjust=0.5, size=10)) +
  theme(plot.subtitle = element_text(hjust=0.5, size=10)) +
  labs(caption = "Figure 3. Vaccination rates for kindergartners",
       x = "Year",
       y = "Vaccination Rates (%) ",
       color = "Type of Vaccine") +
  scale_x_continuous(breaks = c(2009, 2014, 2019, 2024),
                    labels = c("2009", "2014", "2019", "2024")) +
  geom_line()
```

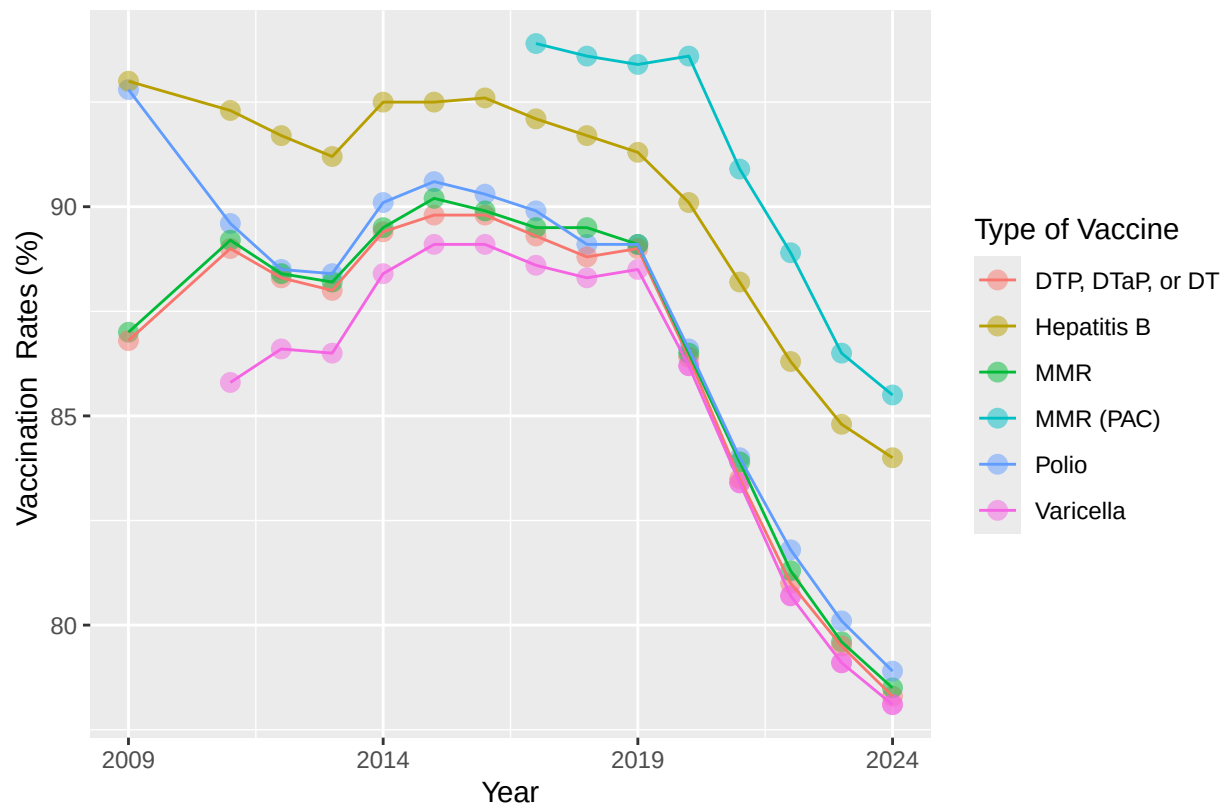


Figure 3. Vaccination rates for kindergartners

5 Data for Social Determinants in Idaho

We found publicly available data on Social Determinants in Idaho (Education levels, Poverty Levels, Median Income, Language Isolation Levels, Voter Affiliations) per county and for the years 2019 to 2023.

```
# loading the data from https://hdpulse.nimhd.nih.gov and Idaho Board of Election

data_poverty <- read.csv("../data/IdahoCounties_Poverty.csv", skip = 4)
data_education <- read.csv("../data/IdahoCounties_Education.csv", skip = 5)
data_medianincome <- read.csv("../data/IdahoCounties_MedianIncome.csv", skip = 4)
data_languageisolation <- read.csv("../data/IdahoCounties_LanguageIsolation.csv", skip = 4)
data_population <- read.csv("../data/IdahoCounties_Population.csv")
data_voters <- read.csv("../data/2025IdahoVoters.csv", skip = 1)

# change class of Voter Counts to Integer
data_voters$CON <- as.integer(
  gsub(",", "", data_voters$CON))
data_voters$DEM <- as.integer(
  gsub(",", "", data_voters$DEM))
data_voters$LIB <- as.integer(
  gsub(",", "", data_voters$LIB))
data_voters$REP <- as.integer(
  gsub(",", "", data_voters$REP))
data_voters$UNA <- as.integer(
  gsub(",", "", data_voters$UNA))
data_voters$Total <- as.integer(
  gsub(",", "", data_voters$Total))
```

Voter Affiliations in Fringe Parties in Idaho

We first look at the distribution of Idaho voter registrations in fringe political parties, such as Libertarian and Constitution, which are known to embrace “extreme” political views. We combine the percentage of voters for the Libertarian Party with those for the Constitution Party and call it the percentage of voters for Fringe Parties.

```
# sum voter registration for fringe parties
data_voters$FRINGE.PERCENT <- data_voters$CON.PERCENT + data_voters$LIB.PERCENT
```

The following histogram looks at “Fringe” Party voters (Libertarian and Constitution affiliated voters are summed). The values of “Fringe” Party voters are pooled in 10 bins for clarity. From the distribution, we can conclude that the rate of Fringe Party Voters does not exceed 2%, hence it is a low subset of the Idaho population, but probably an under representation as one of the tenet of the Libertarian party is to discourage any participation in civic duties such as voting.

```
library(ggplot2)
ggplot(data_voters, aes(x = FRINGE.PERCENT)) +
  geom_histogram(bins = 10,
    fill = "lightblue",
    color = "black") +
  labs(caption = "Figure 4. Distribution of counties with fringe voters",
    x = "% Registered as Fringe Voters",
    y = "number of counties")
```

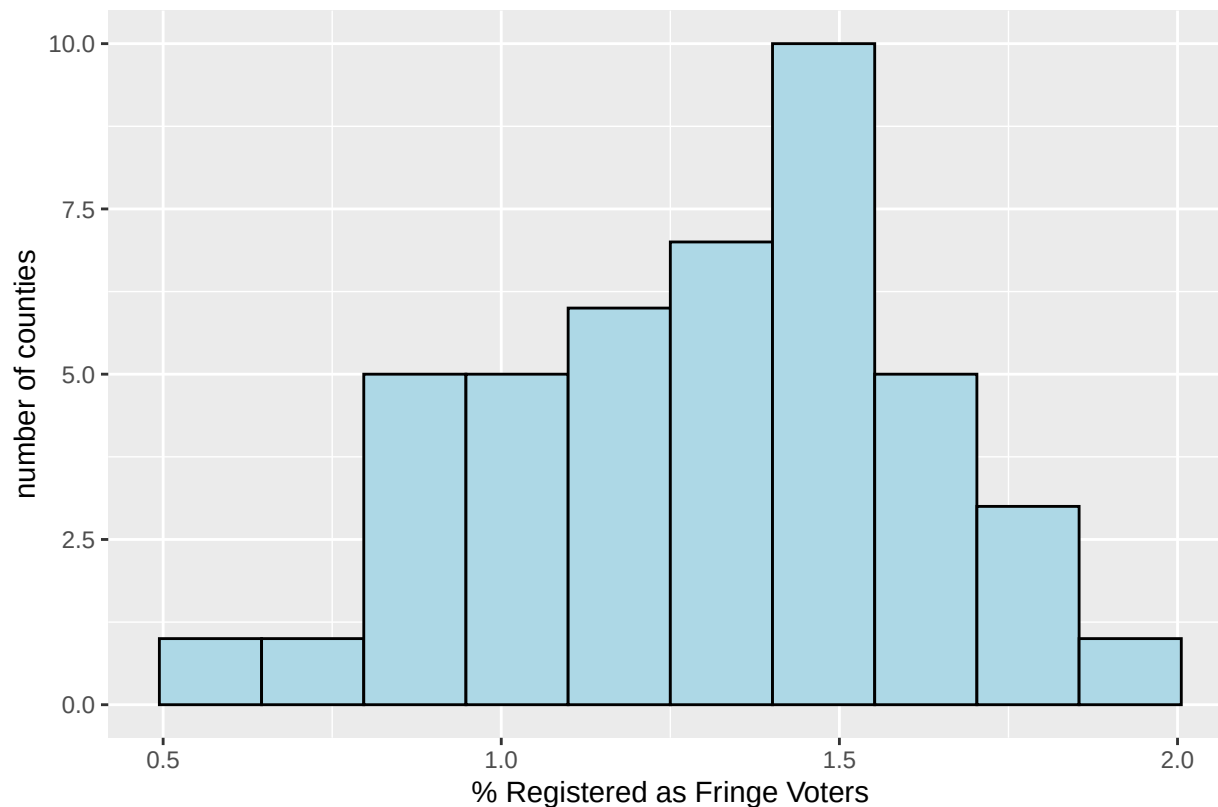


Figure 4. Distribution of counties with fringe voters

```
# county with Max Fringe Voters - which county has the highest value?
max(data_voters$FRINGE.PERCENT)
data_voters$COUNTY[which.max(data_voters$FRINGE.PERCENT)]
data_voters$FIPS[which.max(data_voters$FRINGE.PERCENT)]
```

In order to look at possible correlations between political affiliation, poverty, low education, income, poverty, We merge all the data available for Idaho residents based on variable “FIPS”, which is a numerical code assigned to each county. Because we do not have access to information on individual Idaho residents, we make the assumptions that counties represent a concentration of subjects with a given percentage of a trait and we will compare with other traits observed with a percentage of another trait.

```
library(dplyr)

merged_data <- data_poverty %>%
  full_join(data_education, by = "FIPS") %>%
  full_join(data_medianincome, by = "FIPS") %>%
  full_join(data_languageisolation, by = "FIPS") %>%
  full_join(data_population, by = "FIPS") %>%
  full_join(data_voters, by = "FIPS")

# rename columns
colnames(merged_data)[1] <- "county"
colnames(merged_data)[3] <- "below.poverty.percent"
colnames(merged_data)[7] <- "below.education.percent"
colnames(merged_data)[11] <- "median.income"
```



```

colnames(merged_data)[14] <- "language.isolation.percent"
colnames(merged_data)[19] <- "population.age.18.39"

head(merged_data$median.income)

# change class of median income from character to integer
merged_data$median.income <- as.integer(
  gsub(",", "", merged_data$median.income)
)

```

Determinants of Rural vs Urban residents in Idaho We make the hypothesis that Rural residents in Idaho are more likely to suffer from poverty, low education, and to hold more extreme political views than Urban residents (Westover (2018)). We create a categorical variable “POP” that is either “High” or “Low” for size of population in each county. We set the threshold between “Low” and “High” at 10,000 (data is for population ages 18-39 years). This is not a true representation of the density of population as the counties have different areas. Ideally, we should normalize the number of residents/area, but at this time we make the approximation that each county is of similar area.

```

# create a categorical variable POP
merged_data$POP <- ifelse(merged_data$population.age.18.39 < 10000,
                          "LOW",
                          "HIGH")

```

6 Statistical Analysis for Median Income vs Low/High Population Density

We look at the county median income by categorizing counties by their number of residents in either High or Low density. We can run a t-test to see if the mean of the distributions are significantly different, after checking that both distributions are normal and have similar variances.

```

# check that both distributions are normal

shapiro.test(subset(merged_data, POP == "LOW")$median.income)

```

```

##
## Shapiro-Wilk normality test
##
## data: subset(merged_data, POP == "LOW")$median.income
## W = 0.9439, p-value = 0.09665

```

```

shapiro.test(subset(merged_data, POP == "HIGH")$median.income)

```

```

##
## Shapiro-Wilk normality test
##
## data: subset(merged_data, POP == "HIGH")$median.income
## W = 0.96767, p-value = 0.885

```

```
# calculate Statistical values (mean, median, sd)- verify that sd are similar for both
library(dplyr)
merged_data %>%
  group_by(POP) %>%
  summarise(
    Mean = mean(median.income, na.rm = TRUE),
    Median = median(median.income, na.rm = TRUE),
    SD = sd(median.income, na.rm = TRUE)
  )
```

```
## # A tibble: 2 x 4
##   POP      Mean Median      SD
##   <chr> <dbl> <dbl> <dbl>
## 1 HIGH  84941.  86237 11313.
## 2 LOW   77285.  75666 10273.
```

```
# run t-test
tt <- t.test(merged_data$median.income[merged_data$POP == "LOW"],
             merged_data$median.income[merged_data$POP == "HIGH"])

# display a table with the t-test result
# clean table display

library(knitr)
library(broom)

kable(tidy(tt), digits = 3,
      caption = "Independent Samples t-test Results")
```

Table 2: Independent Samples t-test Results

estimate	estimate1	estimate2	statistic	p.value	parameter	conf.low	conf.high	method	alternative
-	77284.84	84941.17	-2.049	0.055	18.237	-	187.082	Welch Two	two.sided
7656.323						15499.73		Sample t-test	

The t-test shows a p value of 0.05514, hence the null hypothesis cannot be rejected. We conclude that the average of county median incomes of counties of High count of residents is not significantly different to that of counties of Low count of residents.

7 Visual Inspection of the Data for Median Income vs Low/High Population Density

Idaho County Median income distribution, categorized by either Low or High resident count

We pool the county median income data in 8 bins for clearer representation. We superimpose the two histograms obtained for median income for counties for either “High” or “Low” county population.

```
# plotting median income for Idaho with 8 bins of income per county population density

library(ggplot2)

ggplot(merged_data, aes(x = median.income, fill = POP)) +
  geom_histogram(bins = 8, alpha = 0.5, position = "dodge") +
  labs(caption = "Figure 5. Distribution of county median income",
       x = "Median Income",
       y = "Count",
       fill = "County Population")
```

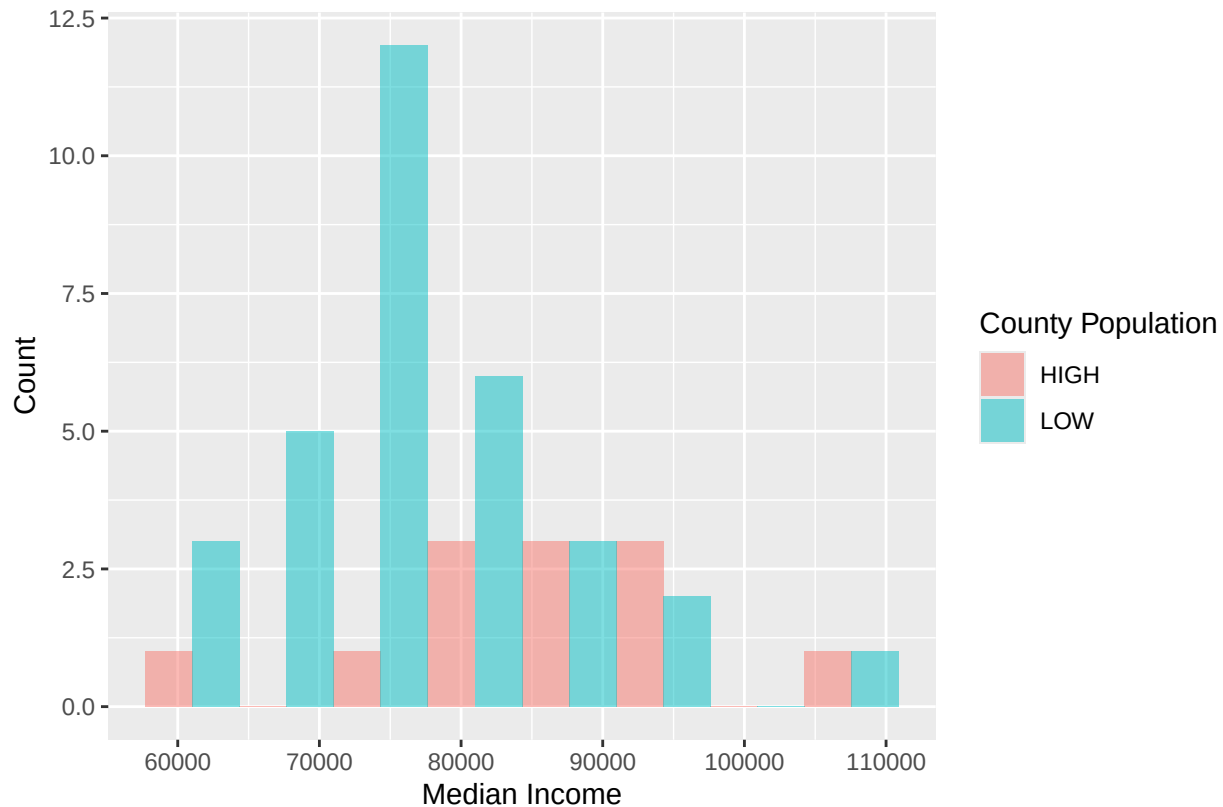


Figure 5. Distribution of county median income

The Distribution of County Median Income shows that the modes of the two distributions are different but that there is some overlap and hence this explains why the p-value corresponds at 94% confidence but fails the 95% confidence that we set by default.

8 Statistical Analysis by Correlation for Median Income vs Poverty Level

As a control, we look at the correlation (and plot) the County Median Income vs the County reported % of Below Poverty. Because we do not have access at the data per resident, we compare averaged data per county. This is a very unrefined approach but we use it a learning tool. But we should see a correlation between county median income and county percentage of below poverty residents.

```
# calculate correlation between Median Income and % below poverty.
cor.test(merged_data$median.income, merged_data$below.poverty.percent)

##
## Pearson's product-moment correlation
##
## data: merged_data$median.income and merged_data$below.poverty.percent
## t = -4.6498, df = 42, p-value = 3.291e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.7500269 -0.3459420
## sample estimates:
## cor
## -0.5829513
```

Results: The Correlation factor is -0.583- the p-value is 3.291e-05, hence as expected there is a significant inverse correlation between median income and % poverty level county-wise, that is a county with a high median income will have a lower percentage of residents below the poverty level. The % below poverty level for Idaho Counties ranges from 2.5 to 19.1%.

9 Visual Inspection of the Data for Median Income vs Poverty Level

The data are represented as a bar plot with the x-axis representing the 4 quartiles of Poverty Levels observed for Counties and the y-axis representing the County Median Income averaged over Counties falling in the same quartile of Poverty Level. Individual Data Points are represented as dots and Standard Deviations are represented as vertical Lines on top of bars.

```
library(ggplot2)
library(Hmisc)
library(dplyr)

# find min and max Poverty levels and which county has the highest poverty level
min(merged_data$below.poverty.percent)

## [1] 2.5

max(merged_data$below.poverty.percent)

## [1] 19.1

merged_data$COUNTY[which.max(merged_data$below.poverty.percent)]

## [1] "BUTTE"

# find the quartiles of the poverty levels observed
quantile(merged_data$below.poverty.percent)
```

```
##      0%    25%    50%    75%   100%
##  2.500  6.475  7.550  9.550 19.100
```

```
# bins the below.poverty.percent variable in quartiles
merged_data <- merged_data %>%
  mutate(poverty_quartile = ntile(below.poverty.percent, 4))

merged_data$poverty_quartile <- factor(
  merged_data$poverty_quartile,
  levels = c(1, 2, 3, 4),
  labels = c("Lowest poverty",
             "Low-middle",
             "High-middle",
             "Highest poverty")
)

# plot county median income vs the 4 quartiles bins of percentage below poverty
ggplot(merged_data, aes(x = poverty_quartile, y = median.income)) +
  stat_summary(fun = mean, geom = "bar", fill = "lightblue") +

# SD error bars
  stat_summary(fun.data = mean_sdl,
              fun.args = list(mult = 1),
              geom = "errorbar",
              width = 0.2) +

# Scatter points (jittered)
  geom_jitter(width = 0.15,
             alpha = 0.5) +

  labs(y = "Averaged county median income",
       x = "Below poverty level quartiles",
       caption= "Figure 6. Averaged median county income by poverty quartile")
```

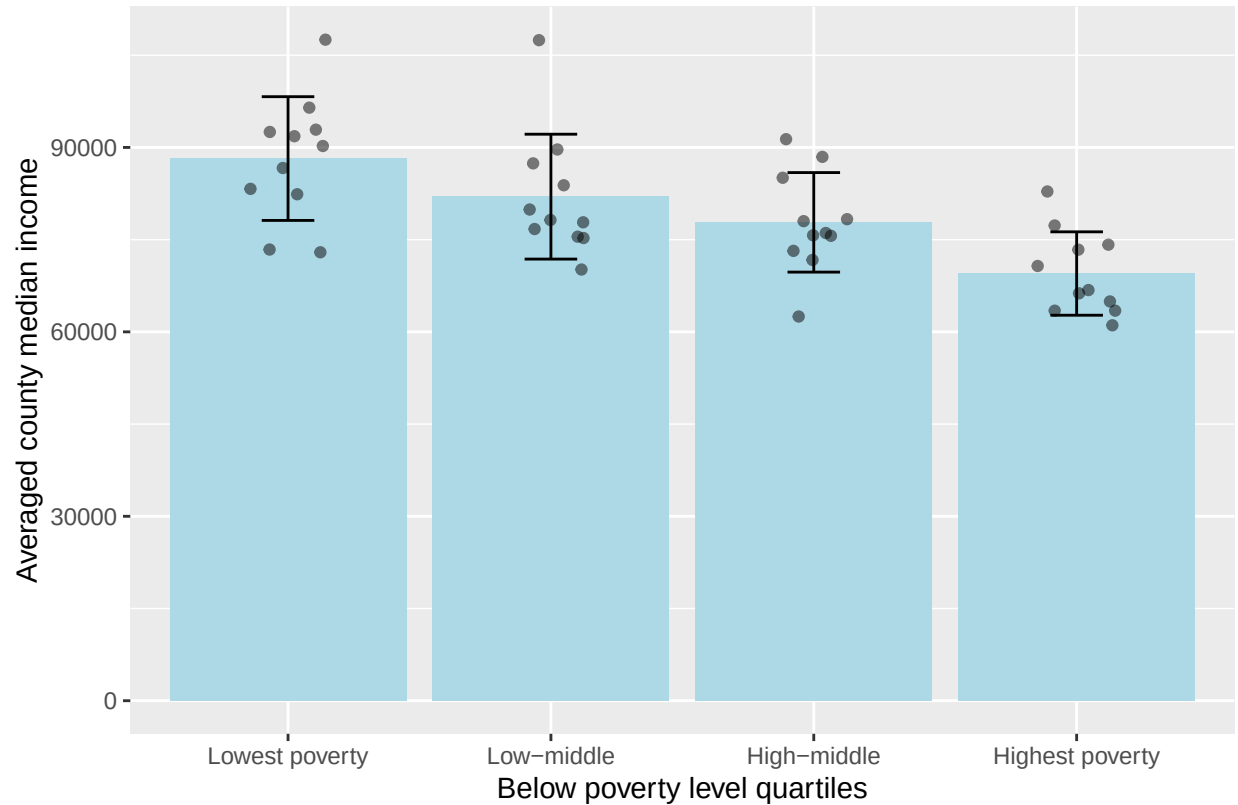


Figure 6. Averaged median county income by poverty quartile

10 Conclusion

This is a work in progress with data that has a lot of noise. But for the sake of the exercise, we can conclude that, at first sight and not surprisingly, there is a clear inverse correlation between County Median Income and County reported % of residents under Poverty Level. We also found that the less densely populated Idaho counties do not have significantly lower Median Income than the more densely populated ones.

11 AI Use Disclosure Statement

As part of this assignment, please indicate whether you used any AI-based tools (e.g., ChatGPT, Claude, Copilot, Gemini, etc.). Indicate:

- I used ChatGPT and old Rmarkdown projects to find the correct functions to use and how to use them. I also resorted to ChatGPT to understand error messages and how to fix them. This document was generated by resorting to the strategy of trial and error, with the use of ChatGPT and past markdown projects to find available libraries, functions and function syntax. The most challenging part of the project is to find usable data.

Literature Cited

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